

Infrastructure regulation in developing countries:

Recent trends in improving regulatory
design and performance

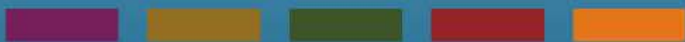
PPIAF Annual Meeting, Bonn, Germany, 23-24 May 2006

Prof Anton Eberhard

Management program in infrastructure reform and regulation
University of Cape Town

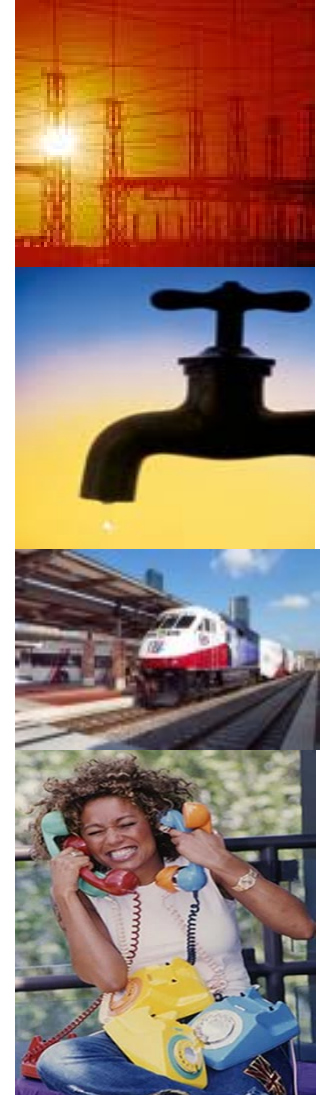


Leading change in emerging markets



Overview

- **Introduction: PPIAF's contribution**
- **Problems with regulatory performance**
- **A menu of regulatory options**
 1. Regulation by agency / Independent regulator
 2. Regulation by contract
 3. Outsourcing regulatory functions
 4. Advisory regulators / expert panels
 5. Regional Regulators
- **Towards improved regulatory design**
 - Best-fit between country context and regulatory options
- **New challenges**



PPIAF's contribution to infrastructure regulation

- Effective regulation was seen as key to infrastructure reform and attracting private investment
- It was hoped that regulation would de-politicise tariff-setting – thus enabling cost-recovery and more predictable revenue streams – while at the same time protecting consumers
- PPIAF has supported at least 165 regulatory activities, totalling US\$ 39.7 m
- Country, regional and global projects
- Focus on institutional design, establishment and capacity building
- PPIAF has also supported important new thinking around regulatory performance and design

Have original expectations been met?
How have regulatory systems performed?

How has PPIAF's role vis a vis regulatory development changed?
What might PPIAF's role be in the future?



African Forum of Utility Regulators

“AFUR recommends that the following key principles form part of an initial framework for utility regulation in Africa:

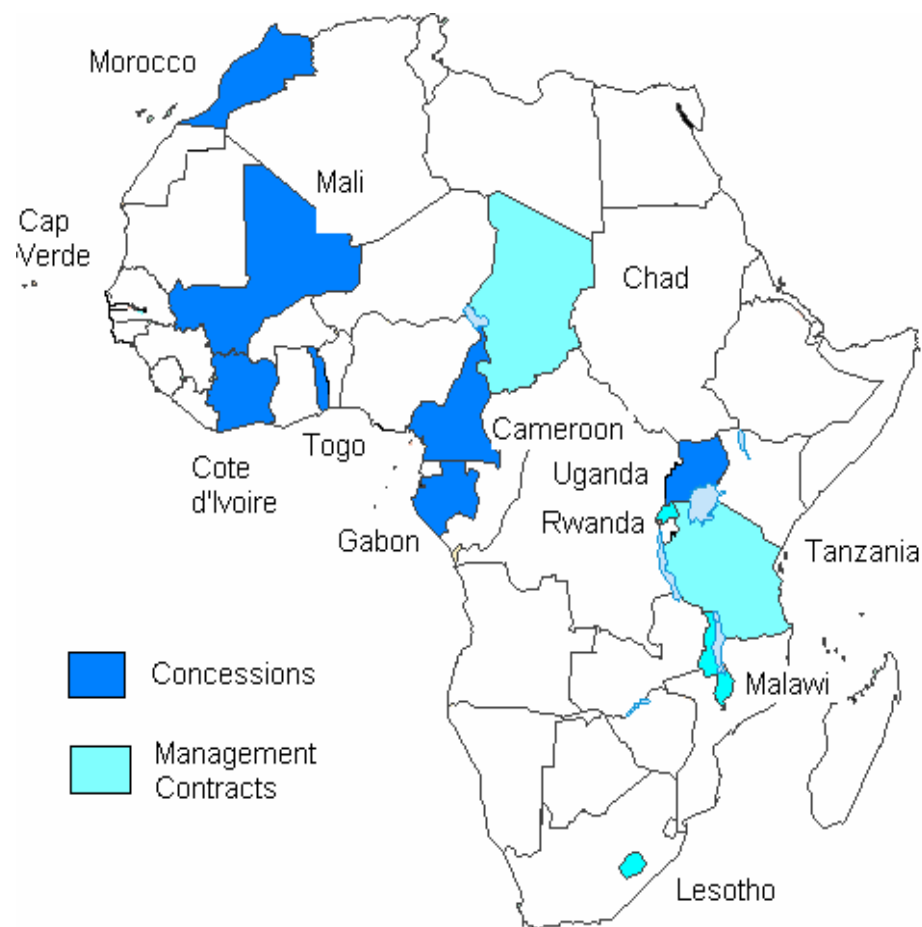
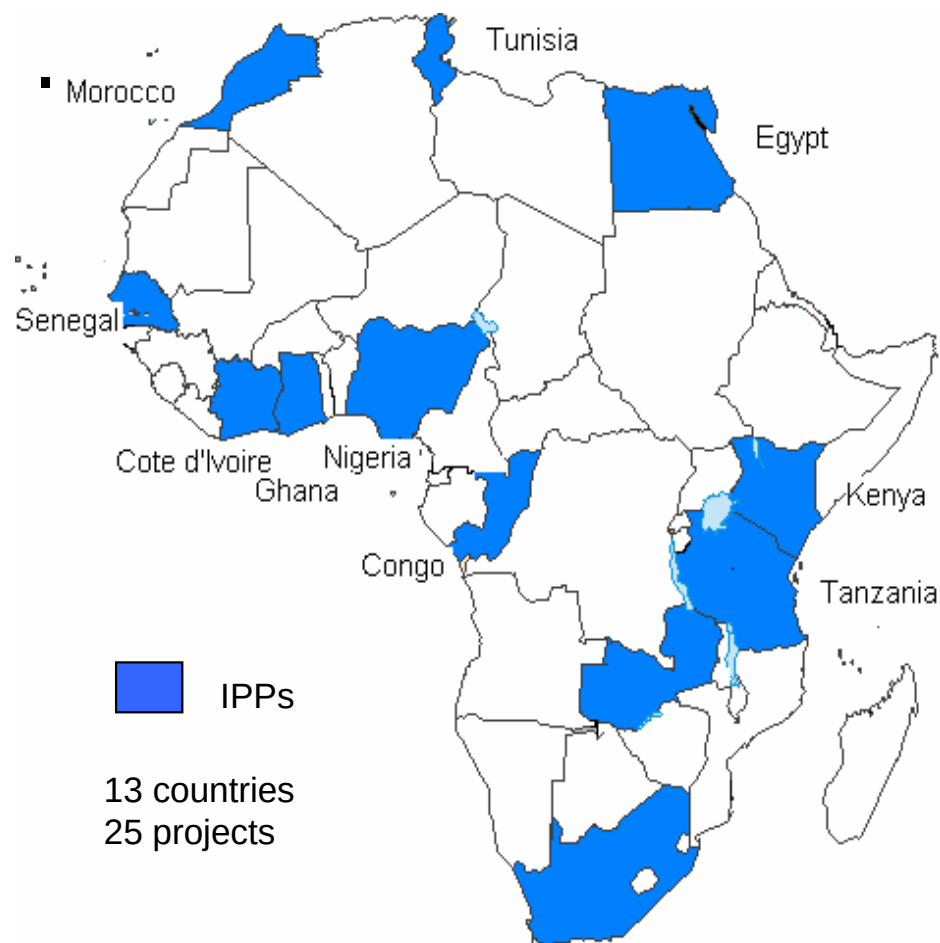
- Minimum regulation necessary to achieve policy and sector objectives;
- Adherence to transparent decision-making and due process requirements;
- **Independent or autonomous regulation where possible;**
- Accountability towards government, investors and end-users
- Non-discrimination when not in conflict with policy prerogatives of government;
- Protection of investors against physical and regulatory expropriation; and
- Promotion of competition by limiting anti-competitive behaviour.”



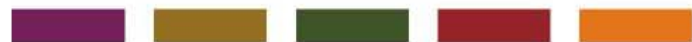
PPIAF supported: A Framework for Utility Regulation in Africa 2003



But also regulatory contracts associated with private participation (e.g. electricity sector in Africa)



Source: PPI database



Developing countries have different regulatory traditions

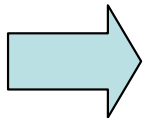
Anglo model

- **Independent regulatory agencies**
- **Considerable regulatory discretion** in setting tariffs and service standards
- But bounded and accountable
- Common law (case law precedents)
- Appeal system often uses ordinary courts

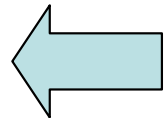
Francophone model

- **Regulatory contracts** (e.g. concession contracts)
- Pre-specified tariff regimes
- **Low levels of discretion**
- Civil law
- Contract renegotiation (e.g. for “adverse government action” or “unexpected constraints”)

Key issue in regulatory design is level of discretion



Increasingly we see hybrids of the two models



Significant regulatory challenges are emerging

- **Limits to independence**

- Some regulators only advise the Minister who makes the final tariff decision
- Political difficulties where prices are not revenue-sufficient
- In many cases, tariff setting is still highly politicized
- Government departments undermine regulators?
- Members of regulator boards are frequently replaced
- Gap between “law” and “practice”

- **Lack of transparency and participation**

- Public explanations for decisions? Stakeholder input?
- Only ¼ of regulators required by law to disclose reasoning
- Transparency often compromised in regulatory contracts
- Transparency most needed where governance & capacity challenges – but least likely in these cases

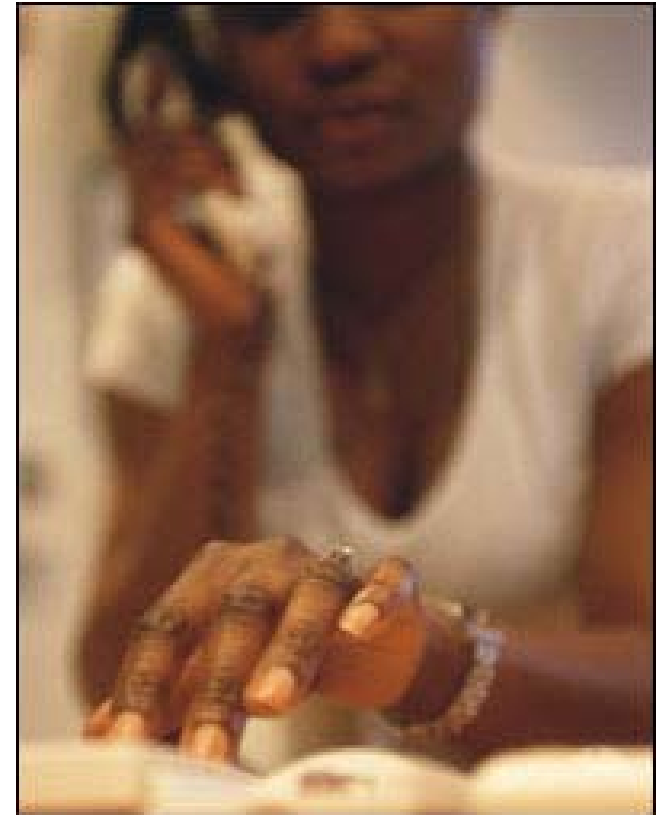
- **Institutional fragility**

- Most regulators only a few years old
- Regulators involved in policy making

- **Lack of capacity/competency**

- Arbitrary or inconsistent decision-making
- “Quality people scarcer than money”
- Greater discretion requires experienced and competent staff

- **A number of regulatory contracts under stress**



Rethinking the classic independent regulator model

THEN

“A requirement of all power lending will be explicit movement toward the establishment of a legal framework and regulatory processes satisfactory to the Bank.....this requires countries to set up transparent regulatory processes that are clearly independent”

The WB's Role in the Electric Power Sector
World Bank Policy Paper 1993

NOW

“.. a credible regulatory system requires more than a formally independent regulatory entity.....other transitional arrangements may need to be established.... including limiting the amount of discretion that regulatory bodies have in setting prices and key parameters..”

Public and private sector roles in the supply of electricity services
Operational Guidance for World Bank Group Staff 2004



A menu of regulatory options

1. Independent regulator
2. Regulation by contract
3. Outsourcing of regulatory functions
4. Strong advisory regulators (expert panels)
5. Regional regulators



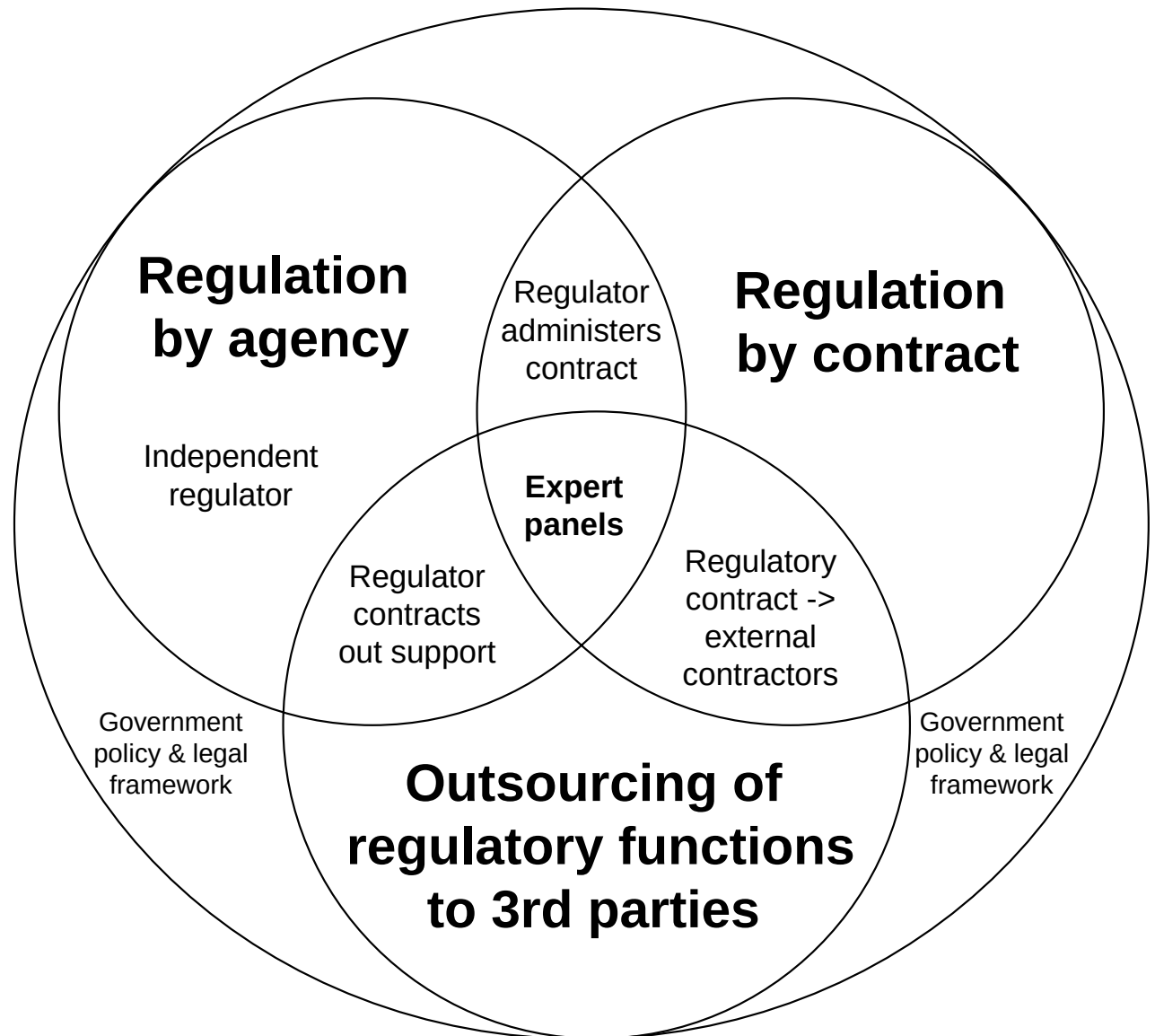
Select from menu
to create transitional or hybrid model



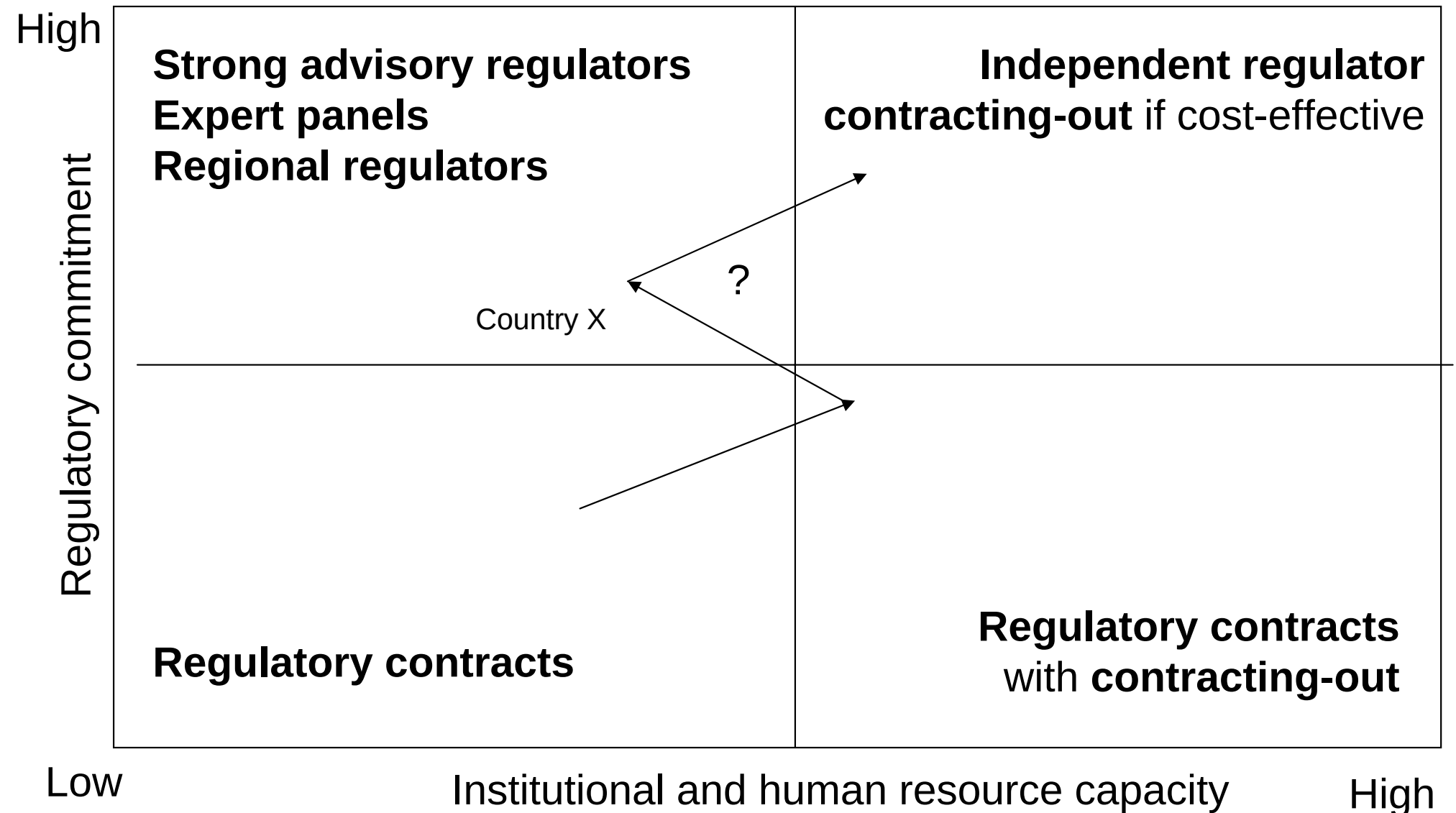
Cf. PPIAF supported studies on these options



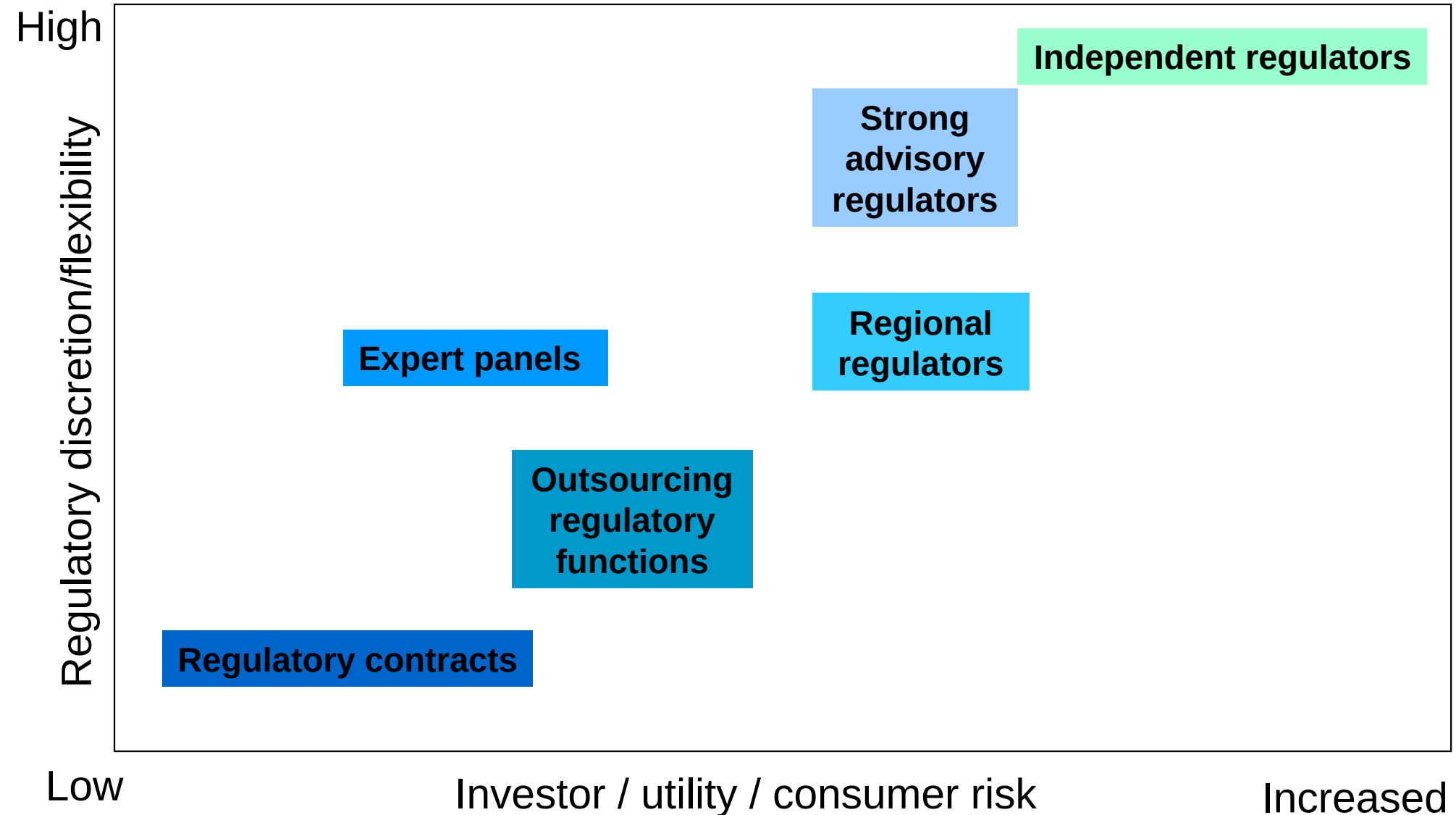
Regulatory design options



Best-fit between country context and regulatory options



Regulatory discretion and risk



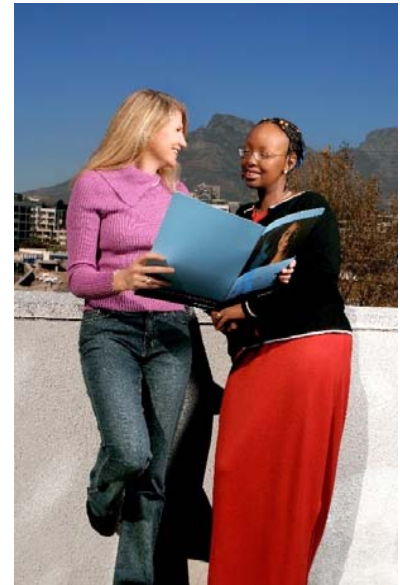
Towards improved regulatory performance

Discussed already

- Choose level of regulatory discretion commensurate with local context of regulatory commitment and capacity

Further measures

- Mandatory, independent reviews of regulators
- Building the demand-side for regulatory transparency and participation
- Capacity building

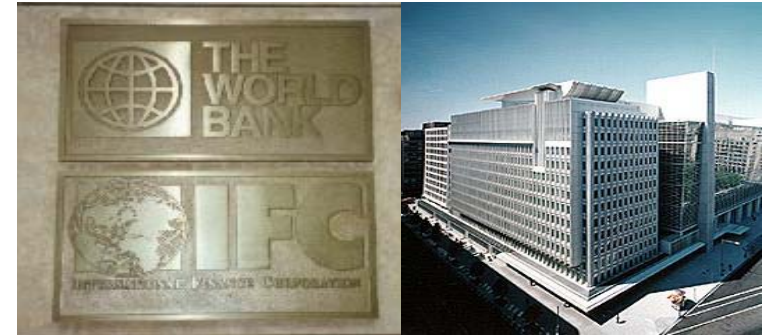


Cf. PPIAF supported capacity building activities, incl establishment of regional orgs.



Partial risk guarantees for regulatory systems

- **First WB regulation PRG - Uganda Electricity Distribution**
- **For loss of regulated revenues from a guaranteed event**
 - Non-compliance by regulator of pre-agreed tariff framework
 - If bulk price not fully passed-through
 - If tariff adjustment is not timely (45 days)
 - If GOU electricity bills are not paid (60 days)
 - If termination payments are not made
- **Provisional PRG payments made pending dispute resolution**
 - Liquidity of utility facilitated
 - Described as “deal-clinching” by investor
- **Regulation PRG considered for Lesotho LEC concession**
- **Also adopted in Romania**



Other risk mitigation measures

- Political risk insurance
- Partial risk guarantees/partial credit guarantees
- Additional financial security measures for investors
 - sovereign guarantees
 - escrow accounts
 - letters of credit
 - stand-by debt facilities
 - hedging and other derivative instruments
 - committed public budget and/or taxes/levies
 - targeted subsidies and output based aid
 - hard-currency contracts
 - indexation in contracts
- Change of law exemption in contracts
- Bilateral Investment Treaties
- Appeal, arbitration and other dispute resolution mechanisms



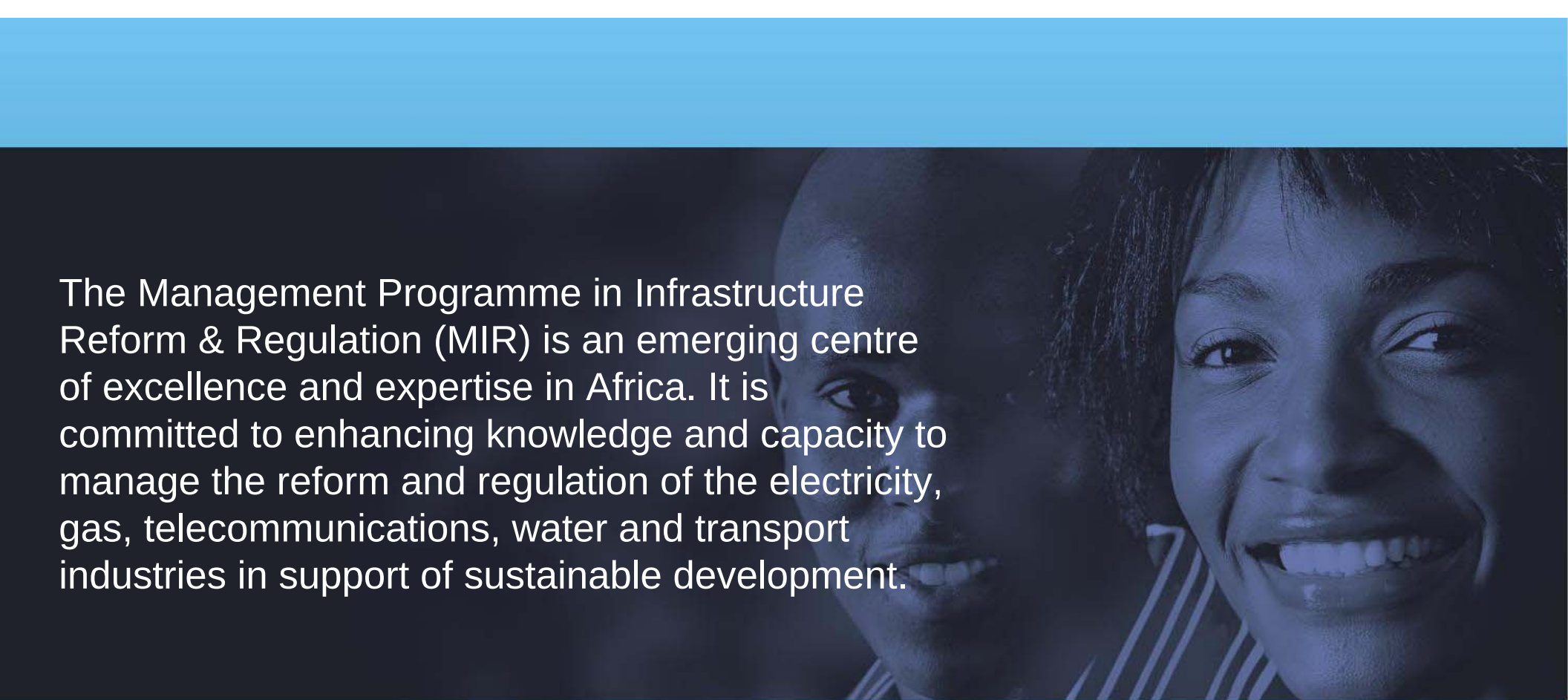
An old (new) challenge: improving the performance of SOEs

- Infrastructure reform in many countries stops short of full competition and privatisation: hybrid markets with state-owned enterprises (SOEs) and private investors are common
- Although focus has been on regulation of private operators – regulation of SOEs is also critical and can play NB role in improving performance, provided that:
 - Government role as political representative and policy maker is separated from its role as utility owner and as regulator
 - SOE is corporatised and commercialised
 - Performance contracts are in sync with regulatory objectives
 - Management are held accountable for performance and sanctions / penalties can be applied
 - Cf work by Prof Gomez-Ibanez



- Effective regulation still seen as key to infrastructure reform and attracting private sector investment
- “Mantra” of independent regulation is being balanced with more sensitively designed regulatory systems that fit local context
- Hybrid and/or transitional regulatory models being explored: independent regulation, regulation by contract, outsourcing, expert panels, regional regulators, etc
- Continued support for institutional development at country level – also key role in establishing and building regional organisations
- **PPIAF continues to play a catalytic role in funding new thinking around regulatory design and performance**





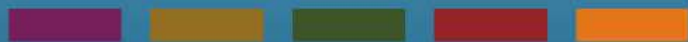
The Management Programme in Infrastructure Reform & Regulation (MIR) is an emerging centre of excellence and expertise in Africa. It is committed to enhancing knowledge and capacity to manage the reform and regulation of the electricity, gas, telecommunications, water and transport industries in support of sustainable development.

Prof Anton Eberhard

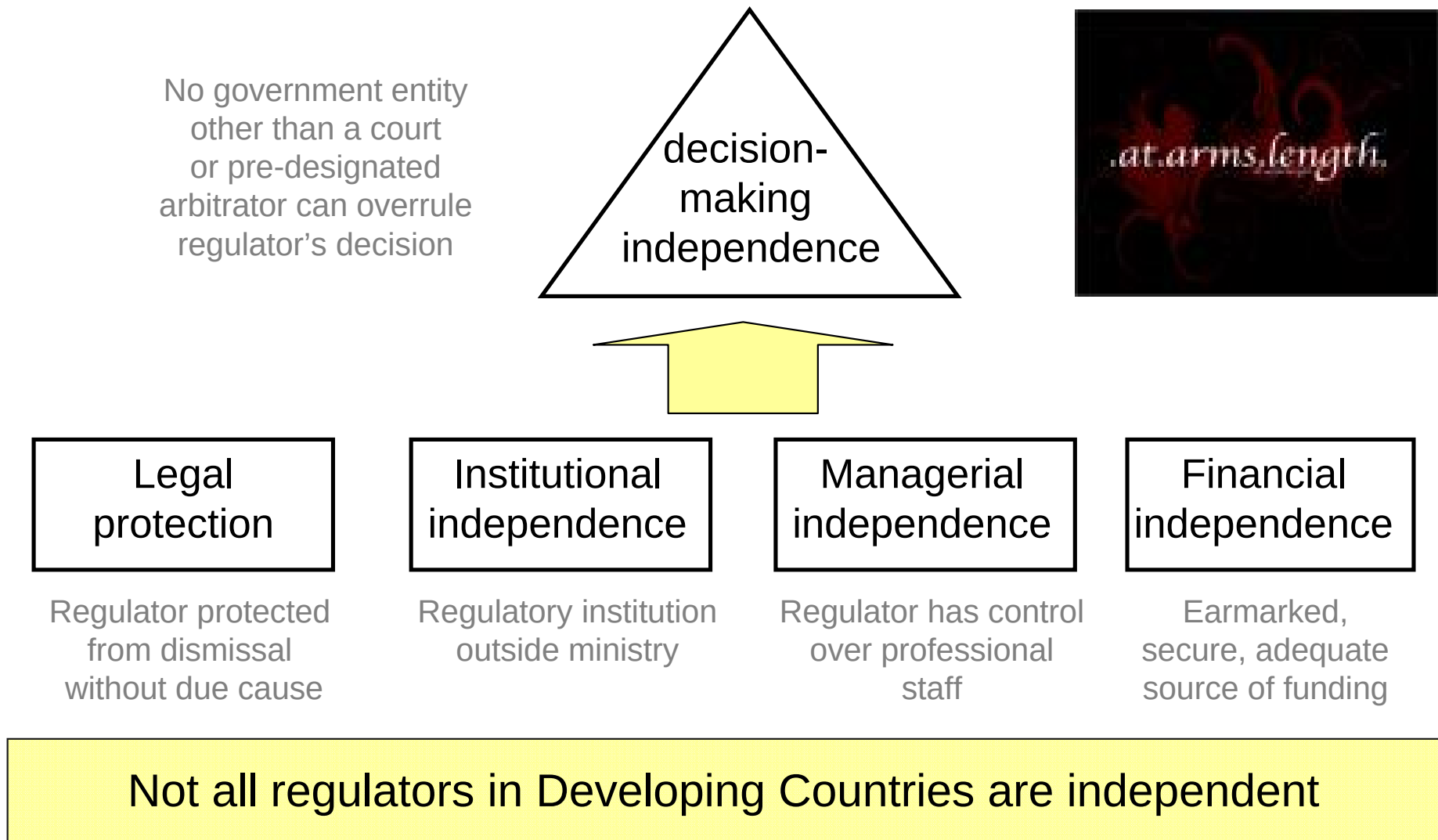
Research, training courses, consultancy
University of Cape Town



Leading change in emerging markets



A1. Regulation by agency / independent regulator



A2. Regulation by contract

A regulatory regime,
including multi-year tariff-setting systems (for 1st period),
pre-specified in one or more legal instrument or contract

- Sometimes approximates the French concession or *affermage* model without the Conseil d'Etat
- Mostly within context of private sector participation but regulatory contracts also possible for state-owned utilities
- Key issues are base-line data; efficiency targets; cost pass-through; new investment; foreign exchange adjustment; triggering events; dispute resolution

Three variants:

1. Key contract provisions self-administered
2. Regulator administers contract
3. Regulatory contract makes provision for outsourcing



A3. Outsourcing of regulatory functions

The use of external contractors or panels either by regulatory agencies or as stipulated in a regulatory contract to perform or assist with certain regulatory functions

- Considered when challenges or problems regarding regulator's independence, competence/capacity, or legitimacy or where regulatory contracts require additional support.
- Strategic decisions needed around core competency and cost-benefits and extent of contracting out
- Potential benefits: increase in regulatory competence, enhanced credibility
- Can be politically sensitive and requires sound contract management

Two broad variants:

1. Consulting support for regulators or regulatory contract
2. Advisory regulators / expert panels

A4. Advisory regulators / expert panels

- **Weak advisory regulator**

- Advice is confidential
- No obligation on Minister to explain rejection or modification of recommendations
- Little or no public consultation
- No separate earmarked budget

- **Strong advisory regulator**

- Regulator's advice must be public
- Ministerial policy directives to regulator must be public
- Regulator has public consultations
- If Minister fails to respond within specified time, recommendations adopted
- Minister must explain publicly rejection or modification of recommendations
- Separate, earmarked funding
- Pre-scheduled, periodic regulatory assessments
- Role could be specified in primary or secondary legislation

Expert panel could substitute for independent regulator / regulatory contract
or could be used in support: e.g. arbitration



Brown, Stern, Tenenbaum & Gencer 2006



A5. Regional regulators

- In principle.....
 - Certain decisions on pricing and interconnection of regional networks could be contracted-out to a regional regulator.
- In practice.....
 - Governments and regulators are unwilling to cede “sovereignty” (telecoms an exception?)
 - Regional institutions face huge challenges in terms of political commitment, institution building and resources
- One option is to create regional expert panels to support national regulators



A6. Mandated periodic assessments of regulators

Pre-scheduled, periodic, in public domain

- Mandated in primary or secondary legislation
- *Ex post* evaluation, includes recommendations
- Should cover both regulatory governance and regulatory substance
- Should include impact of regulator's actions and decisions on sector outcomes
- Performed by a panel of independent national and international experts
- Hasn't been mandated anywhere in Africa, but examples of ad-hoc assessments
- Rating / benchmarking across regions?

A7. Building the demand-side for regulatory transparency^{p6}

The best guarantee for ensuring legitimate, credible and transparent regulatory institutions is to build demand among stakeholders



Brown, Stern, Tenenbaum & Gencer 2006



Slide 25

p6

Maybe this is a pre-condition for regulation and should be put up front first, before anything else.

As I see it, the demand will drive the type of mechanisms that will be chosen.

If stakeholder demand is upfront, then it is possible to discuss why different stakeholder groups would demand regulation.

paul.sotkiewicz, 11/02/2005

A8. Capacity building

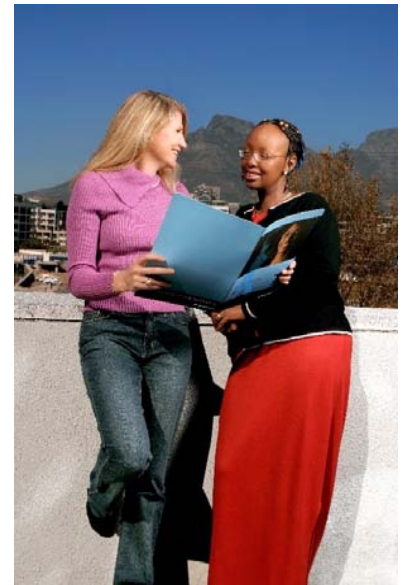
Credibility, quality and impact of regulatory decisions depends in large measure on competence of regulatory staff

“We lack good people”...

”Quality human resources are scarcer than money...”
- African regulator

Train, train, train.....

- Fly-in, once-off regulatory training courses with standard curricula and “northern” regulators as teachers can be ineffective.
- Regional training and education centres can provide sustainable capacity building that is relevant to African needs while bringing in international expertise when warranted



Tremolet & Shah, 2005



A9. Evaluating regulatory systems

Regulatory governance

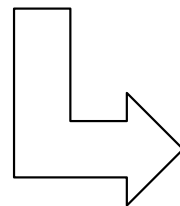
Legal design and institutional arrangements of regulatory system and processes of regulatory decision-making

Credibility, legitimacy, and transparency of regulatory decisions

Regulatory substance

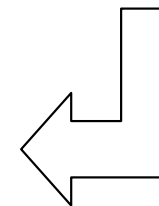
Content of regulation
tariff-setting
service standards

Quality & robustness of regulatory decisions



Impact on sector

Cost-effective, reliable infrastructure services
Financial viability, new investment



Adapted from Brown, Stern, Tenenbaum & Gencer, 2006



A10. Summary

- An effective **regulatory** system needs robust regulatory **substance** and credible, legitimate and transparent regulatory **governance**
- Enhancing regulatory **substance** implies improving the **competence** of regulatory staff
- Regulatory **governance** structures and processes should **constrain arbitrary** administrative **action**
 - Key question: how much regulatory discretion?
- Success of a regulatory system depends on **compatibility** with country's regulatory commitment and institutional and human resource endowment
- Select from a menu of regulatory options to create **hybrid model**
- Nature of hybrid model can change over time as regulatory independence and capacity is built
 - a **transitional model**

